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COMPUTER SCIENCE DEPARTMENT

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Computer Networks & Data Communication

Project Report

Design and Implementation of a Financial Institution System Networking Project

Jubilee Financial Services Ltd (JFSL)

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Chapter 1:

Abstract

Jubilee Financial Services Ltd (JFSL) is a well-established financial services organization operating from Nairobi, Kenya. Due to increasing operational demands, security concerns, and the need for high availability, the company decided to design and own its complete network infrastructure. This project presents the design and implementation of a secure, scalable, and redundant enterprise network using Cisco Packet Tracer.

The network follows a hierarchical design model and supports multiple departments distributed across two floors of the headquarters, with an external server-side site connected through a WAN. Key technologies implemented include VLANs, Inter-VLAN Routing, OSPF dynamic routing, DHCP, DNS, HTTP, Email servers, NAT/PAT, site-to-site IPsec VPN, VoIP telephony services, wireless networking, and Access Control Lists (ACLs) for security. The final network ensures confidentiality, integrity, availability, and reliable communication between all departments and the remote server site.

Chapter 2: Introduction

2.1 Background

Jubilee Financial Services Ltd operates within an eleven-storey building, utilizing the seventh and eighth floors for its headquarters operations. The organization consists of five main departments: Human Resources (HR), Customer Service (CS), Marketing (MK), Legal Management (LM), and Information Technology (IT). Each department requires both data and voice communication, secure access to centralized servers, wireless connectivity, and internet access.

Previously, the network infrastructure was managed by a third-party provider. Senior management decided to internalize network ownership to improve security, performance, redundancy, and scalability while maintaining full control over LAN, WAN, and server-side communication.

2.2 Project Objectives

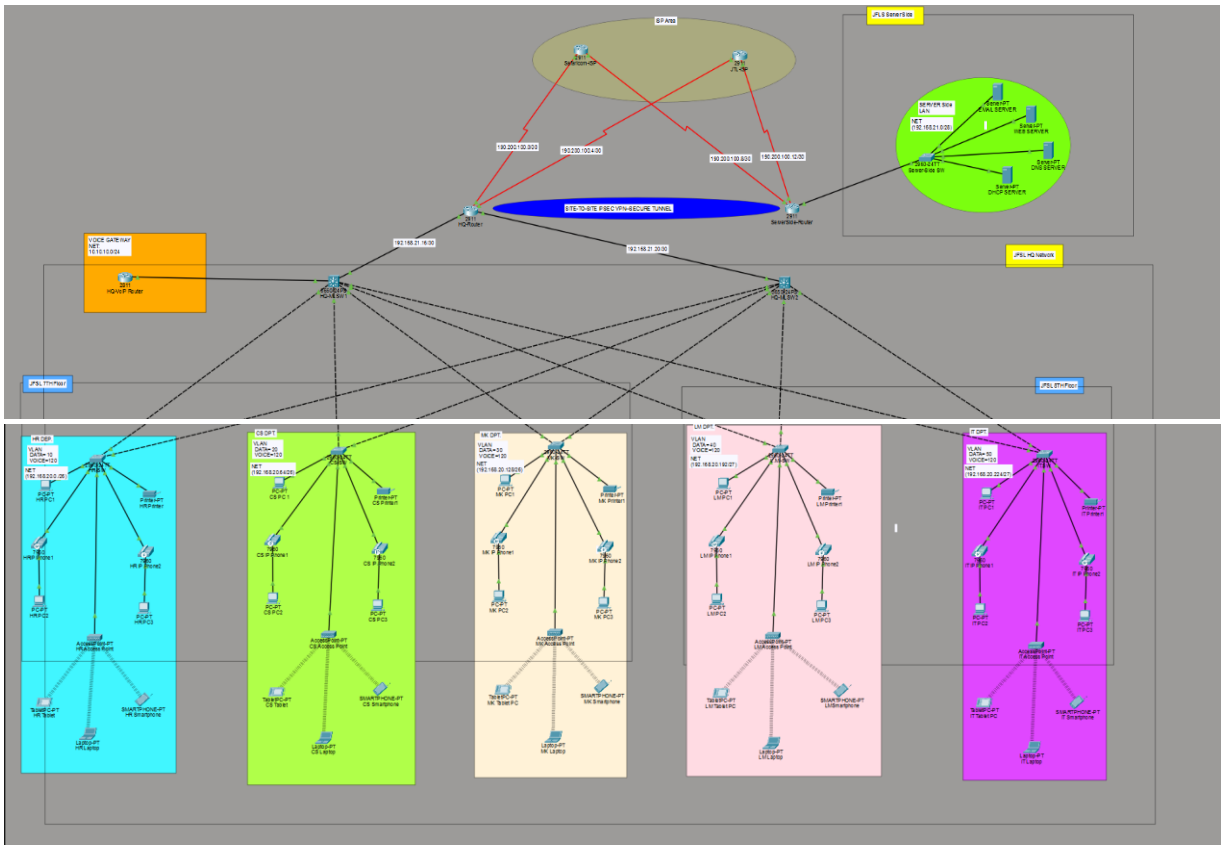
The main objectives of this project are:

- To design a hierarchical enterprise network with redundancy at all layers
- To implement VLAN-based segmentation for all departments
- To configure inter-VLAN routing using multilayer switches
- To deploy secure WAN connectivity between HQ and server-side site
- To implement OSPF as the dynamic routing protocol
- To configure centralized servers (DHCP, DNS, WEB, EMAIL)
- To enable VoIP communication across departments
- To enforce security using ACLs, NAT/PAT, VPN, port security, and SSH
- To test and verify full network functionality

Chapter 3: Methodology

3.1 Network Topology & Design Plan

The network configuration simulated in Cisco Packet Tracer for the **Jubilee Financial Services Ltd (JFSL) Network Design Project** adheres to Cisco's hierarchical network model, emphasizing high performance, scalability, redundancy, and secure communication. The design is structured into three distinct layers: the core layer, distribution layer, and access layer. At the core layer, redundancy and high availability are achieved through the deployment of two multilayer switches that perform both routing and switching functions, along with Cisco 2911 routers responsible for headquarters and server-side routing. The distribution layer consists of departmental access switches that connect and segment each department into separate Virtual Local Area Networks (VLANs) to enhance traffic management and security. The access layer accommodates end-user devices, including desktop computers, IP phones, and wireless access points, providing both wired and wireless connectivity. Additionally, the network integrates redundant Internet Service Provider (ISP) links from Safaricom and JTL to ensure uninterrupted internet access and load balancing. A secure site-to-site IPsec Virtual Private Network (VPN) is implemented to connect the headquarters network with the external server-side site, ensuring confidential and reliable data communication. This hierarchical topology results in a well-organized, resilient, and manageable network infrastructure capable of supporting the organization's current operations and future expansion.



3.2 Components

3.2.1. Routers (4):

- Two Cisco Catalyst 2911 routers are deployed for headquarters (HQ) and server-side site routing.
- One Cisco Catalyst 2811 router is dedicated as the VoIP gateway to support telephony services.
- Routers are positioned at the core layer to provide redundancy, WAN connectivity, and secure routing.
- Each router is connected to two Internet Service Providers (Safaricom and JTL) to ensure high availability and load balancing.
- Routers are configured with static public IP addresses provided by the ISPs and support NAT, OSPF, SSH, and IPsec VPN services.

3.2.2. Multilayer Switches (2):

- Deployed at the core layer of the headquarters network to provide redundancy and high-performance routing.
- Configured to perform both Layer 2 switching and Layer 3 routing functionalities.
- Assigned IP addresses and configured with Switch Virtual Interfaces (SVIs) to enable inter-VLAN routing.
- Participate in OSPF routing to exchange routes with routers and other network devices.

3.2.3. Distribution Layer Switches (Multiple):

- Dedicated access switches are deployed for each department across the seventh and eighth floors.
- These switches connect individual departments to the core multilayer switches.
- Facilitate secure communication within respective data and voice VLANs.
- Configured with access and trunk ports, along with VLAN and voice VLAN assignments.

3.2.4. End-User Devices (PCs, Laptops, Printers, Phones and Tabs):

- Deployed at the access layer within all departments.
- Connected to distribution layer switches for departmental network access.
- Obtain IP addresses dynamically from the centralized DHCP servers located at the server-side site.

3.2.5. Cisco Wireless Access Points (APs):

- Positioned at the access layer in each department.
- Provide wireless connectivity for staff and authorized users.

- Integrated with departmental VLANs to ensure secure wireless access.

3.2.6. DHCP Server (1):

- Located at the external server-side site.
- Responsible for dynamically allocating IP addresses to all end-user devices across the headquarters network.
- Configured to supply default gateway and DNS server information to clients.

3.2.7. Server-Side Devices (Servers and Infrastructure):

- Includes dedicated **DNS Server (192.168.21.6)** and **WEB Server (192.168.21.7)**, along with other supporting servers such as Email services.
- All server-side devices are assigned static IP addresses for consistency and reliability.
- These servers provide essential services such as domain name resolution, web hosting, and centralized network services.
- Server-side devices are securely connected to the headquarters network via a site-to-site IPsec VPN.

These devices collectively form a structured, secure, and well-organized network architecture that supports high performance, redundancy, scalability, and secure communication in line with the operational requirements of Jubilee Financial Services Ltd (JFSL).

3.3 IP Addressing Scheme

Details about the IP addressing scheme applied to the network:

3.3.1. Base Network:

192.168.20.0/24

3.3.2 Department classification

Department	Network	Subnet Mask	Default Gateway	VLAN
HR	192.168.20.0/26	255.255.255.192	192.168.20.1	10
CS	192.168.20.64/26	255.255.255.192	192.168.20.65	20
MK	192.168.20.128/26	255.255.255.192	192.168.20.129	30
LM	192.168.20.192/27	255.255.255.224	192.168.20.193	40
IT	192.168.20.224/27	255.255.255.224	192.168.20.225	50

3.3.3 Voice Network

- Network: 10.10.10.0/24
- Voice VLAN ID: 120

3.3.4 Server-Side LAN

- Network: 192.168.21.0/28
- Default Gateway: 192.168.21.1

3.3.5 Core Router and L3 SW

- HQ Router - Multilayer Switch 1: 192.168.21.16 /30
- HQ Router - Multilayer Switch 2: 192.168.21.20 /30

3.3.6 Public IP between Core and ISP

- HQ Router- Safaricom ISP: 190.200.100.0/30
- HQ Router - JTL ISP: 190.200.100.4/30
- Server-Side Router - Safaricom ISP: 190.200.100.8/30
- Server-Side Router - JTL ISP: 190.200.100.12/30

3.4 Routing Configuration

Details about the configuration of different devices used in network.

3.4.1 Router Configuration

Routers were configured with:

- **Basic settings (hostname, passwords, banners)**

```
en
conf t

hostname HQ-

enable password cisco

line console 0
password cisco
login
exit


banner motd #NO UNAUTHORISED ACCESS!!!!#

no ip domain lookup

service password-encryption
```

By using the same commands, the all core layer and ISP Routers configured to basic settings after changing the name of Router in commands.

- **SSH for secure remote access**

```
username cisco password cisco
ip domain-name cisco.com

crypto key generate rsa
1024

ip ssh version 2

line vty 0 15
login local
transport input ssh
exit
```

- **NAT/PAT for internet access**

```
(config)#int range gig0/0-1
(config-if-range)#ip nat in
(config-if-range)#ip nat inside
(config-if-range)#exit
(config)#
```

For NAT inside, we configure the ports of router that connected to the ML switches or to inside the network.

```
(config)#int se0/2/0
(config-if)#
(config-if)#
(config-if)#ip nat out
(config-if)#ip nat outside
(config-if)#int se0/2/1
(config-if)#ip nat outside
(config-if)#exit
(config)#
(config)#
```

For NAT outside, the ports configured of router which are connected to the outside Network (ISP's).

```
(config)#access-list 50 per
(config)#access-list 50 permit 192.168.20.0 0.0.0.255
(config)#
(config)#
(config)#acc
(config)#access-list li
(config)#ip na
(config)#ip nat
(config)#ip nat in
(config)#ip nat inside s
(config)#ip nat inside source ac
(config)#ip nat inside source access
(config)#ip nat inside source lis
(config)#ip nat inside source list 50 int
(config)#ip nat inside source list 50 interface se0/2/0 ov
(config)#ip nat inside source list 50 interface se0/2/0 overload
(config)#ip nat inside source list 50 interface se0/2/1 overload
(config)#
```

It is applied PAT on the router.

- **OSPF routing protocol**

```
(config)#router os
(config)#router ospf 10
(config-router)#rout
(config-router)#router-id 2.2.2.2
(config-router)#net
(config-router)#network 10.10.10.0 0.0.0.255 ar
(config-router)#network 10.10.10.0 0.0.0.255 area 0
(config-router)#
(config-router)#network 192.168.21.20 0.0.0.3 area 0
(config-router)#
(config-router)#
(config-router)#network 192.168.20.0 0.0.0.255 area 0
(config-router)#exit
(config)#
(config)#
(config)#
```

In this step the configuration of routing protocol will be done. Every device in this configuration get a unique router-id and take values of different paths connected to it. This is mandatory in a same network no device will get same id.

This configuration will be done on all layer three switches all routers available in network.

```
HQ-Router(config)#
HQ-Router(config)#do sh ip ospf neigh

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/DR         00:00:37    192.168.21.21   GigabitEthernet0/1
1.1.1.1           1   FULL/DR         00:00:34    192.168.21.17   GigabitEthernet0/0
6.6.6.6           0   FULL/  -        00:00:39    190.200.100.6   Serial0/2/1
5.5.5.5           0   FULL/  -        00:00:37    190.200.100.2   Serial0/2/0
HQ-Router(config)#
```

It can be checked that all interface connected to device are configured to the routing protocol.

- **Telephony service for VoIP**

```
(config)#
(config)#interface FastEthernet0/0
(config-if)#no ip add
(config-if)#
(config-if)#
(config-if)#
(config-if)#interface FastEthernet0/0.120
(config-subif)#
5-CHANGED: Interface FastEthernet0/0.120, changed state

ROTO-5-UPDOWN: Line protocol on Interface
ernet0/0.120, changed state to up

(config-subif)#
(config-subif)#
(config-subif)#enc
(config-subif)#encapsulation d
(config-subif)#encapsulation dot1Q 120
(config-subif)#ip add 10.10.10.1 255.255.255.0
(config-subif)#exit
(config)#
(config)#
(config)#do wr
ng configuration...
```

To configure voice service on HQ-VoIP router we first perform these steps. We firstly create a separate VLAN for it.

3.4.2 Switch Configuration

Multilayer and access switches were configured to support:

- **Basic settings (hostname, passwords, banners)**

```
en
conf t

hostname CS-SW

enable password cisco

line console 0
password cisco
login
exit


banner motd #NO UNAUTHORISED ACCESS!!!!#

no ip domain lookup

service password-encryption

do wr
```

By using the same commands, the all access layer and multilayer switches configured to basic settings after changing the name of department in commands.

- **SSH for secure remote access**

```
username cisco password cisco
ip domain-name cisco.com

crypto key generate rsa
1024

ip ssh version 2

line vty 0 15
login local
transport input ssh
exit
```

These commands will be applied on all switches

- **VLAN creation for each department**

```
vlan 20
name DATA
```

VLAN created for each department by same procedure on access layer switch just by changing the number of VLAN according to assigned in table

- **Voice VLAN (VID 120)**

```
vlan 120
name VOICE
exit
```

These commands will be applied on all access layer switches to configure the VOICE VLAN.

- **Trunk ports**

```
int range fa0/1-2
switchport mode trunk
exit
```

For **Access layer switches** trunk ports are the interfaces connected to the Multilayer Switches and may change the interface number in commands as the interfaces are connected to Multilayer Switches.

```
(config)#int range gig1/0/2-6
(config-if-range)#swit
(config-if-range)#switchport m
(config-if-range)#switchport mode tr
(config-if-range)#switchport mode trunk
(config-if-range)#exit
(config)#
```

```
vlan 10
name HR
vlan 20
name CS
vlan 30
name MK
vlan 40
name LM
vlan 50
name IT
vlan 120
name VOICE
exit
```

For **Multilayer Switches** trunk Ports are the ports which are connected to the Access layer switches with the VLAN number and name of VLAN of Access Layer Switch connected to Multilayer Switch.

- **Access Ports**

```
int range fa0/3-24
switchport mode access
switchport access vlan 20
switchport voice vlan 120
exit
```

The remaining ports on **Access Layer Switches** left after trunk ports will be assigned to the access mode ports so that they can access the VLAN data to their connected devices.

- **DHCP helper addresses**

```
(config)#int vlan 10
(config-if)#
CHANGED: Interface Vlan10, changed state to up

TO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

(config-if)#
(config-if)#
(config-if)#
(config-if)#ip add 192.168.20.1 255.255.255.192
(config-if)#
(config-if)#ip help
(config-if)#ip helper-address 192.168.21.5
(config-if)#exit
(config)#
(config)#
```

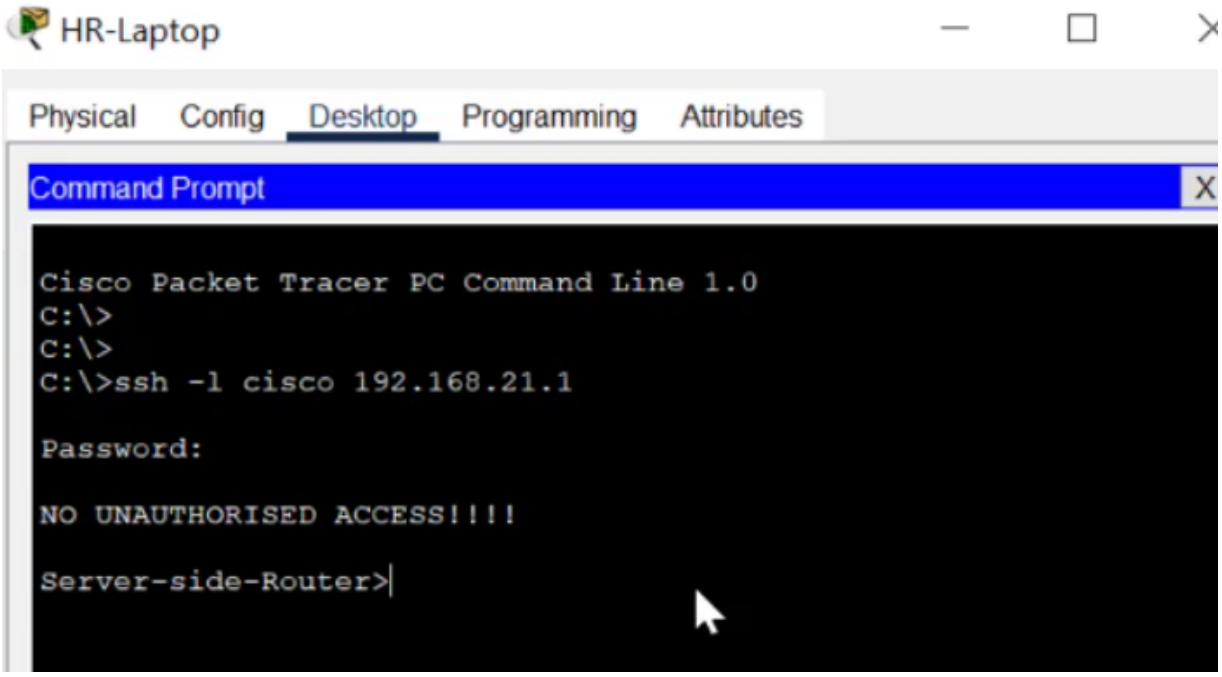
This is provided to Multilayer Switches so that the VLAN's can get IP's from DHCP server as the server able to communicates the networks.

For each VLAN the number of VLAN and IP address is changed in commands according to department and the Helper-Address will be same

3.5 Security Implementation

Security mechanisms implemented include:

- **Standard ACL on VTY lines allowing SSH access only from IT department**



Before implementation of this security protocol the all devices connected inside the network able to configure the layer three and service providing devices which is not good for security. To overcome that apply an ACL so that only IT department can access only that devices.

```
access-list 10 permit 192.168.20.224 0.0.0.31
access-list 10 deny any

line vty 0 15
access-class 10 in
exit

do wr
```

Apply these commands to establish an ACL to all ML switches and Routers, so that no device outside the IT department can access these devices for configuration.

```
C:\>
C:\>
C:\>ssh -l cisco 192.168.20.1

% Connection refused by remote host
C:\>
```

After applying that ACL, the connection is limited to the IT department and not allowed to access by any other department.

IT-PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l cisco 192.168.21.1

Password:
```

This shows the IT department devices can access only the routers and ML switches.

- **Port security on server-side switch (sticky MAC, maximum 1, shutdown)**

```
(config)#
(config)#int range fa0/2-5
(config-if-range)#
(config-if-range)#
(config-if-range)#swit
(config-if-range)#switchport m
(config-if-range)#switchport mode ac
(config-if-range)#switchport mode access
(config-if-range)#
(config-if-range)#
(config-if-range)#swir
(config-if-range)#swit
(config-if-range)#switchport p
(config-if-range)#switchport port
(config-if-range)#switchport port-security
(config-if-range)#switchport port-security max
(config-if-range)#switchport port-security maximum 1
```

Making that sure one port Available only for maximum1 device at a time.

```
SERVER-SIDE-SW(config)#do sh port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
                (Count)       (Count)        (Count)
--------------------------------------------------------------------
        Fa0/2        1          0                0         Shutdown
        Fa0/3        1          0                0         Shutdown
        Fa0/4        1          0                0         Shutdown
        Fa0/5        1          0                0         Shutdown
--------------------------------------------------------------------
```

If any violation occur the port goes automatically shutdown

```
(config-if-range)#switchport port-security mac
(config-if-range)#switchport port-security mac-address st
(config-if-range)#switchport port-security mac-address sticky
(config-if-range)#
(config-if-range)#
(config-if-range)#switchport port-security vi
(config-if-range)#switchport port-security violation sh
(config-if-range)#switchport port-security violation shutdown
(config-if-range)#exit
(config)#
(config)#
(config)#
(config)#do wr
```

Taking Mac-addresses of connected devices so if a strange/unauthorized device trying to connect the port go shutdown.

```
!
interface FastEthernet0/2
 switchport mode access
 switchport port-security
 switchport port-security mac-address sticky
!
interface FastEthernet0/3
 switchport mode access
 switchport port-security
 switchport port-security mac-address sticky
!
interface FastEthernet0/4
 switchport mode access
 switchport port-security
 switchport port-security mac-address sticky
!
```

This is that how connected devices checked by their Mac-address to maintain security.

```
(config)#int range fa0/6-24, gig0/1-2
(config-if-range)#switc
(config-if-range)#switchport mod
(config-if-range)#switchport mode a
(config-if-range)#switchport mode access
(config-if-range)#swit
(config-if-range)#switchport ac
(config-if-range)#switchport access v
(config-if-range)#switchport access vlan 99
does not exist. Creating vlan 99
(config-if-range)#shut
```

Shutting down all the remaining ports not in use so no device connects to empty ports.

```
interface FastEthernet0/6
 switchport access vlan 99
 switchport mode access
 shutdown
!
interface FastEthernet0/7
 switchport access vlan 99
 switchport mode access
 shutdown
!
interface FastEthernet0/8
 switchport access vlan 99
 switchport mode access
 shutdown
!
interface FastEthernet0/9
 switchport access vlan 99
 switchport mode access
 shutdown
!
```

This is that after shutting down ports all vacant ports got down and connected to a VLAN that not exists in the system

- **Site-to-site IPsec VPN between HQ and server-side routers**

This s important to do that the data traveling through the ISP's cant be readable by anyone if trying to sniff the packets.

First we create a tunnel for VPN between HQ and server-side Router

```
(config)#license b
(config)#license boot m
(config)#license boot module c
(config)#license boot module c2900 t
(config)#license boot module c2900 technology-package s
(config)#license boot module c2900 technology-package securityk9
```

Enabling the security license so that the VPN protocols can be applied.

```
access-list 110 permit ip 192.168.20.0 0.0.0.255 192.168.21.0 0.0.0.15


crypto isakmp policy 10
encryption aes 256
authentication pre-share
group 5
exit
crypto isakmp key vpnpa55 address 190.200.100.9
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
crypto map VPN-MAP 10 ipsec-isakmp
description VPN connection to SERVERSIDE-SITE.
set peer 190.200.100.9
set transform-set VPN-SET
match address 110
```

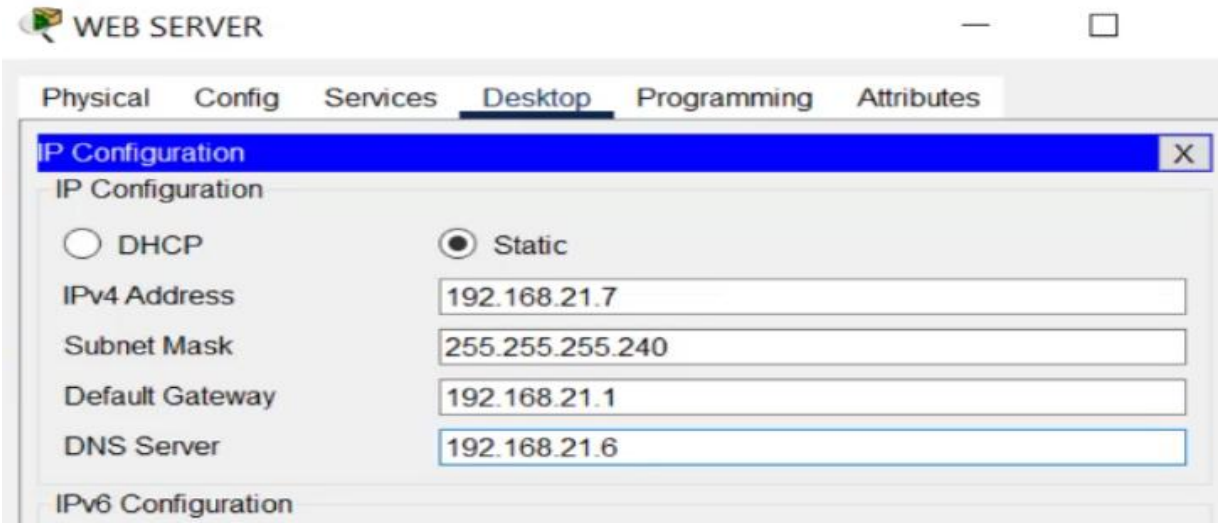
```
exit
interface S0/2/0
crypto map VPN-MAP
EXIT


DO WR
```

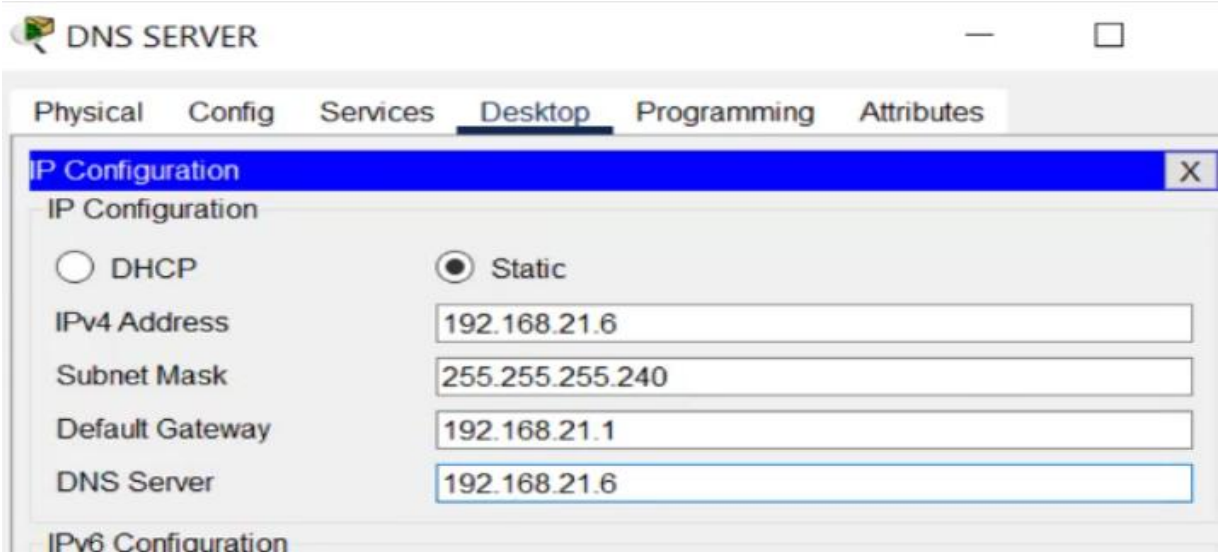
Apply these commands on the Server-Side router and HQ-Router by just changing the IP addresses according to the interfaces to enable VPN service.

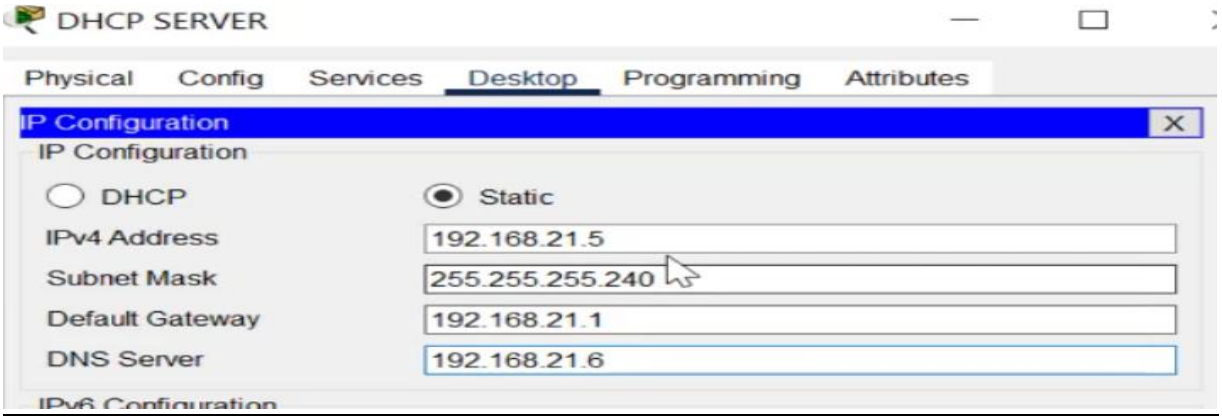
3.7 Server Configuration

The server-side site hosts the following servers:

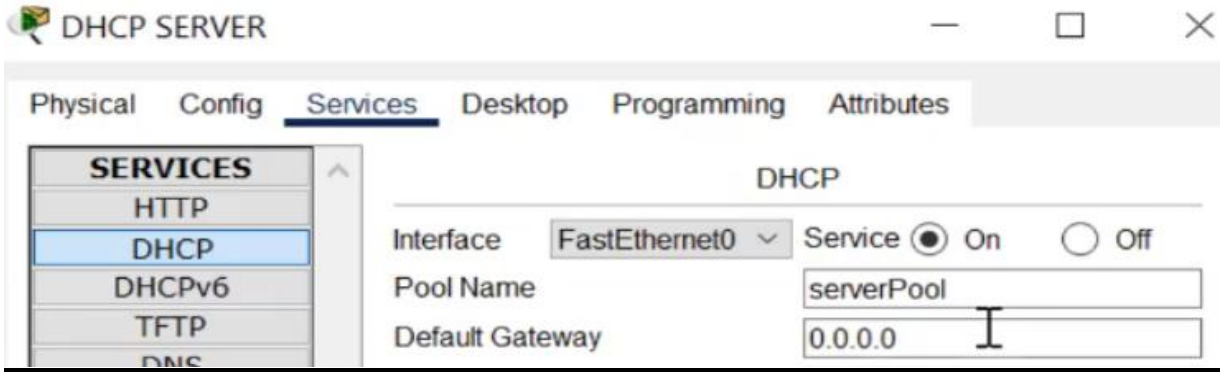
- **All servers are configured with static IP addresses.**



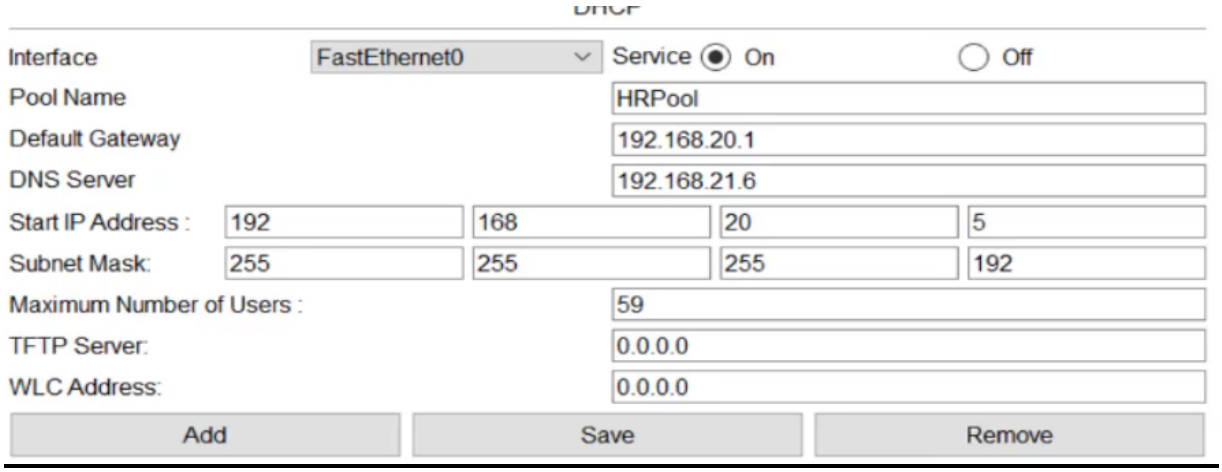




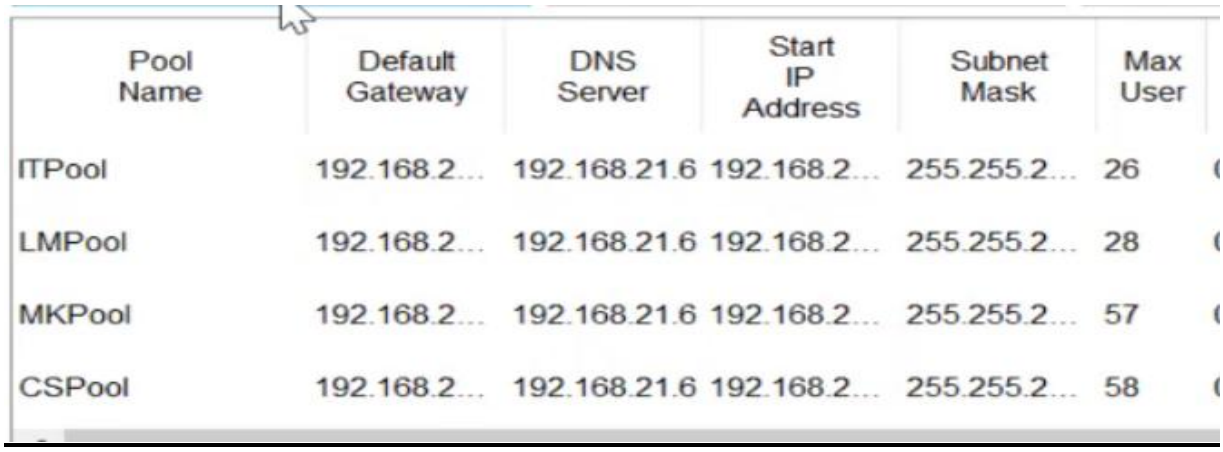
- **DHCP Server (Dynamic IP allocation)**



First we enable DHCP service on DHCP server.

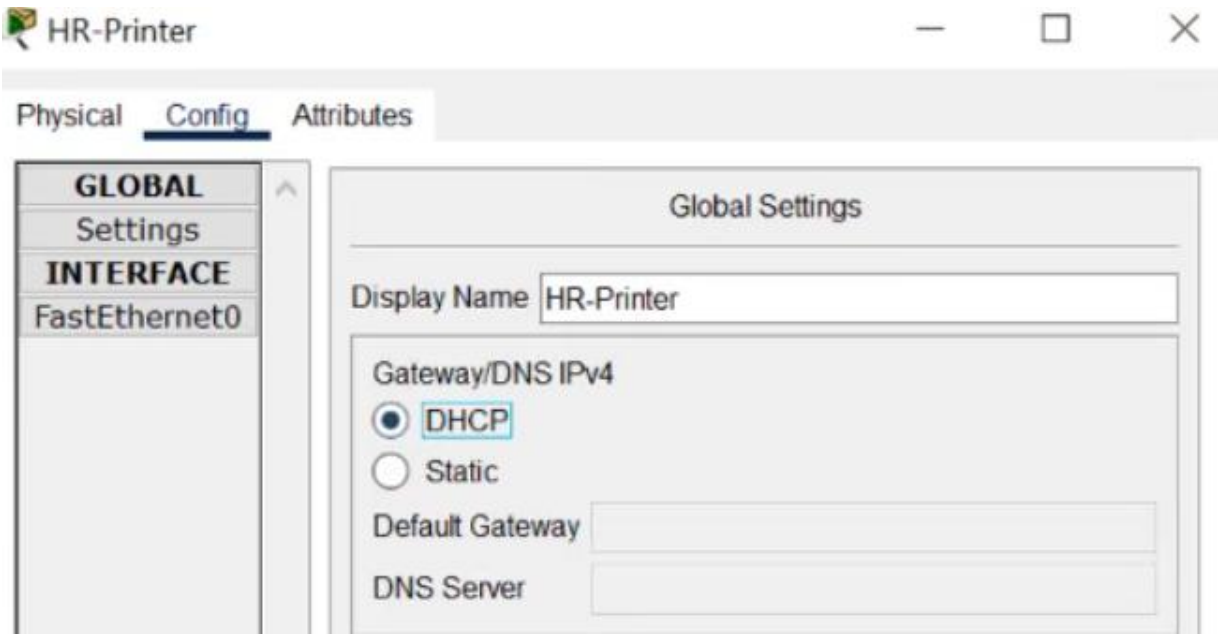


Then we create a pool for each department according to their IP ranges so that server assign IP's in their range.



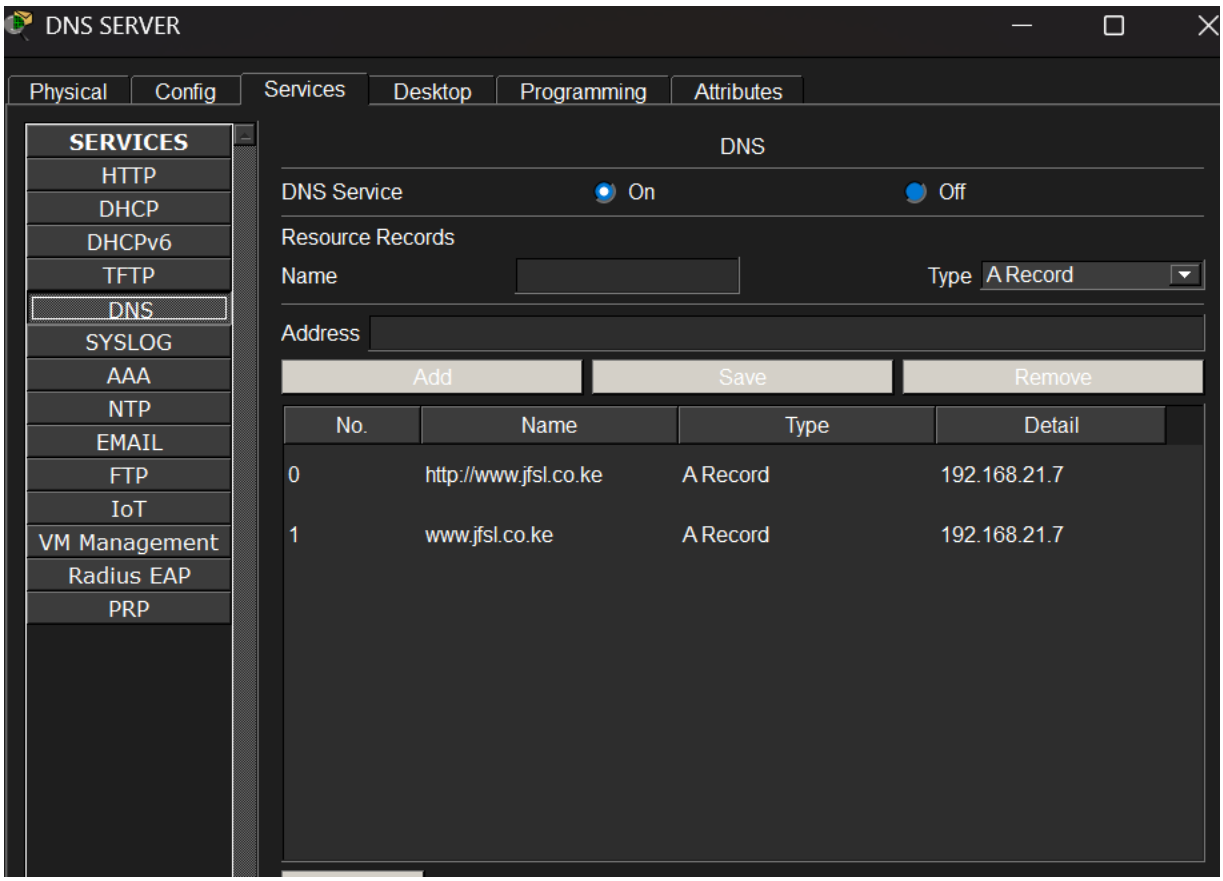
Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User
ITPool	192.168.2...	192.168.21.6	192.168.2...	255.255.2...	26
LMPool	192.168.2...	192.168.21.6	192.168.2...	255.255.2...	28
MKPool	192.168.2...	192.168.21.6	192.168.2...	255.255.2...	57
CSPool	192.168.2...	192.168.21.6	192.168.2...	255.255.2...	58

Pools for different departments in this organization



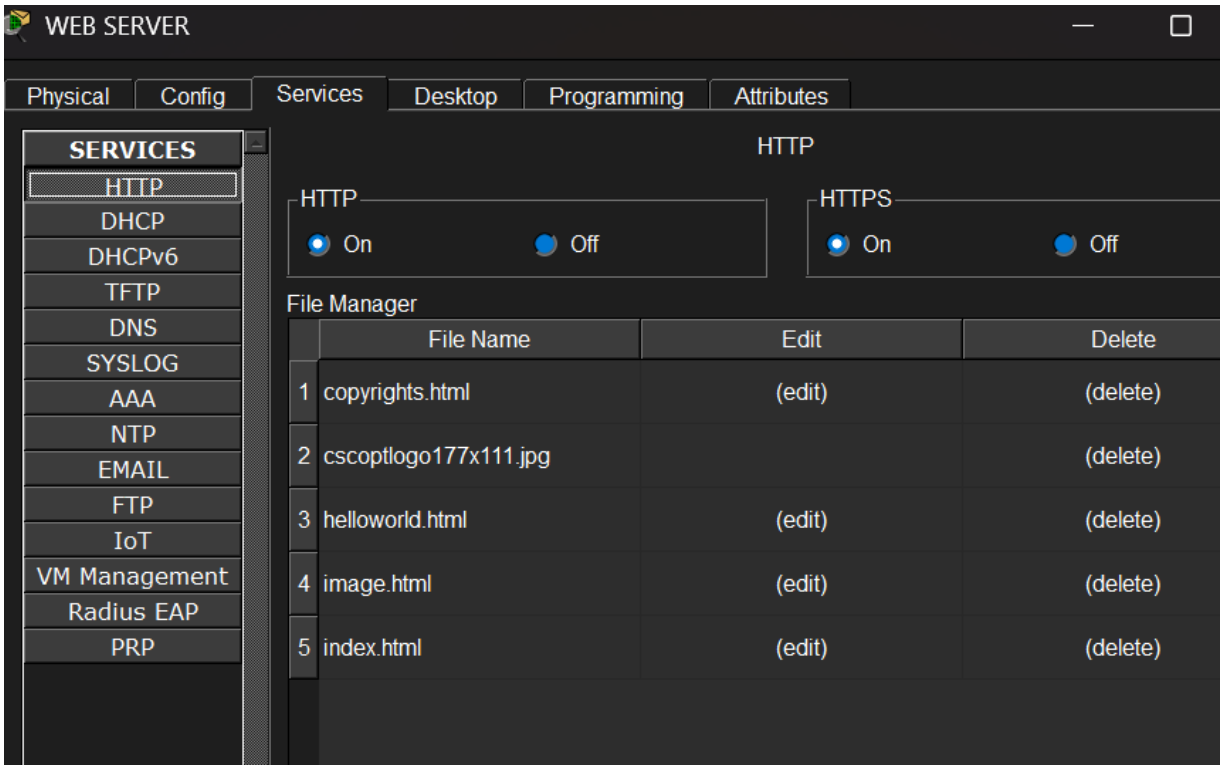
After that, enable the DHCP service on devices connected to network.

- **DNS Server (Domain name resolution)**



Here the DNS server configured with the website name www.jfsl.co.ke or http://www.jfsl.co.ke and address "192.168.21.7" the address of Web Server.

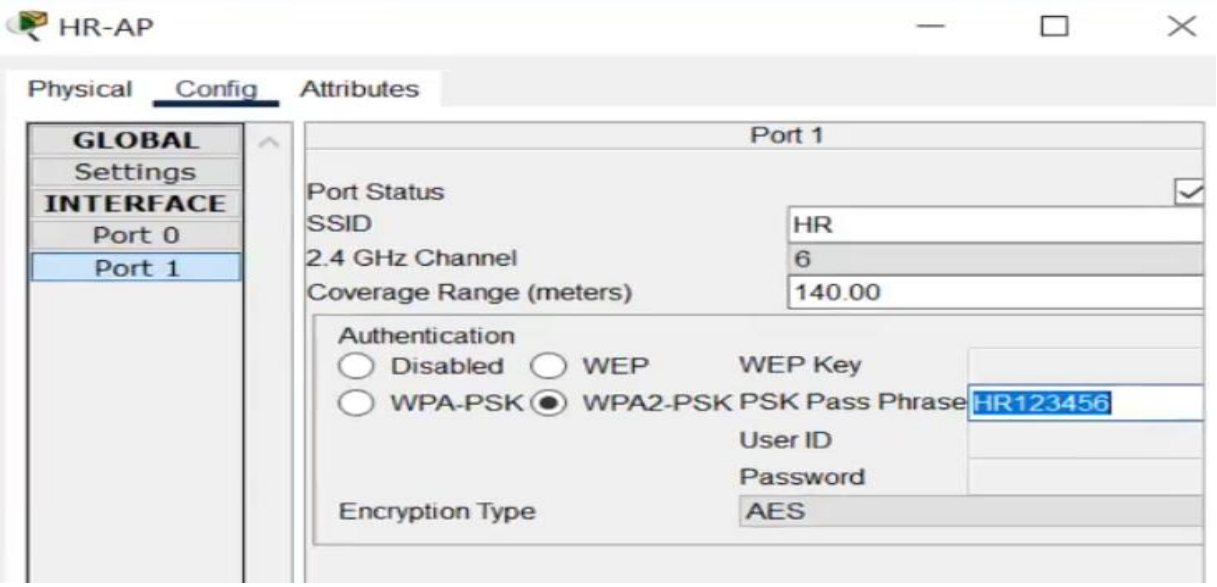
- **HTTP Server (Sample webpage hosting)**



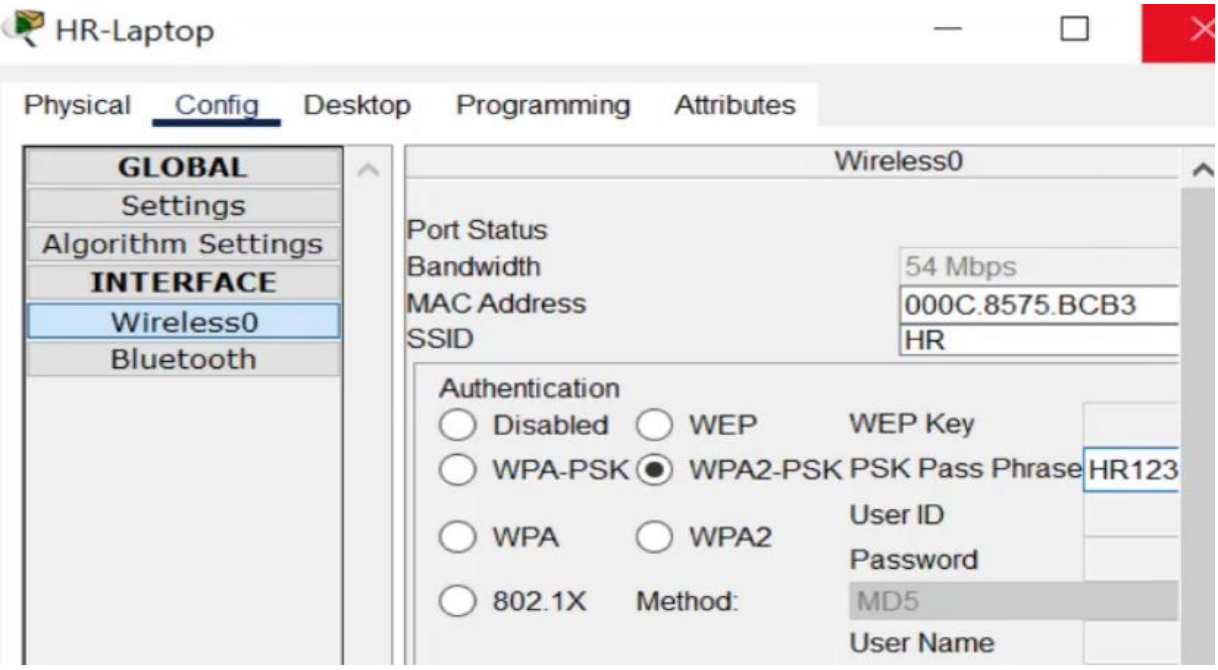
Enable the HTTP services and place a sample webpage which is accessed by the systems of network.

3.8 Wireless & VoIP Configuration

- **Each department has a wireless access point configured with appropriate VLANs and security.**



The Wireless access points also configured by a unique name and a password, so the wireless devices can also connect to the network get IP's through DHCP server.



The devices connect to the access point if the put the correct SSID and Password provided by Access Point.

- **VoIP services were configured on the Cisco 2811 router, with dial numbers assigned in the format (4XX). IP phones receive IP addresses dynamically.**

```
(config)#service d
(config)#service dhcp
(config)#
(config)#
(config)#ip dhc
(config)#ip dhcp po
(config)#ip dhcp pool VOICE
(dhcp-config)#netw
(dhcp-config)#network 10.10.10.0 255.255.255.0
(dhcp-config)#def
(dhcp-config)#default-router 10.10.10.1
(dhcp-config)#
(dhcp-config)#opt
(dhcp-config)#option 150 10.10.10.1
                         ^
lid input detected at '^' marker.

(dhcp-config)#option 150 ?
Data is one or more IP addresses
(dhcp-config)#option 150 ip 10.10.10.1
(dhcp-config)#

(dhcp-config)#dns
(dhcp-config)#dns-server 10.10.10.1
(dhcp-config)#exit
(config)#
```

First enable the DHCP service on router so that the IP phones can get IP's dynamically.

```
(config)#ip dhcp excluded-address 10.10.10.1
(config)#
```

Making sure that the reserved IP address can't be assigned by the DHCP server to telephones while assigning the IP's.

```
(config)#
(config)#teleph
(config)#telephony-service
(config-telephony)#
(config-telephony)#max-e
(config-telephony)#max-ephones 20
(config-telephony)#max-
(config-telephony)#max-dn
(config-telephony)#max-dn 20
(config-telephony)#
(config-telephony)#ip sour
(config-telephony)#ip source-address 10.10.10.1 por
(config-telephony)#ip source-address 10.10.10.1 port 2000
(config-telephony)#aut
(config-telephony)#auto-a
(config-telephony)#auto ass
(config-telephony)#auto assign 1 to 20
(config-telephony)#exit
(config)#
```

After the IP allocation, Telephony service enable to the router. It will allocate the numbers to telephones connected inside the network. At this stage it is also configured that how many telephony devices will be inside the network.

```
(config)#ephone-d
(config)#ephone-dn 1
(config-ephone-dn)#%LINK-3-UPDOWN: Interface ephone_dsp DN 1.1,

(config-ephone-dn)#
(config-ephone-dn)#numb
(config-ephone-dn)#number 401
(config-ephone-dn)#ex
```

At this stage after the allocation of IPs we can now allocate the directories by the number format which we want. In this network the requirement is in 4.. format, So 1’st device get 401 and so on.

Chapter 4: Results

The following tests were performed:

- **Ping tests between VLANs**

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	LM PC1	IT PC1	ICMP		0.000	N	6	(edit)	(delete)
	Successful	HR PC2	CS PC2	ICMP		0.000	N	7	(edit)	(delete)
	Successful	CS PC2	MK PC2	ICMP		0.000	N	8	(edit)	(delete)
	Successful	HR PC2	LM PC2	ICMP		0.000	N	9	(edit)	(delete)

- **Connectivity to server-side site**

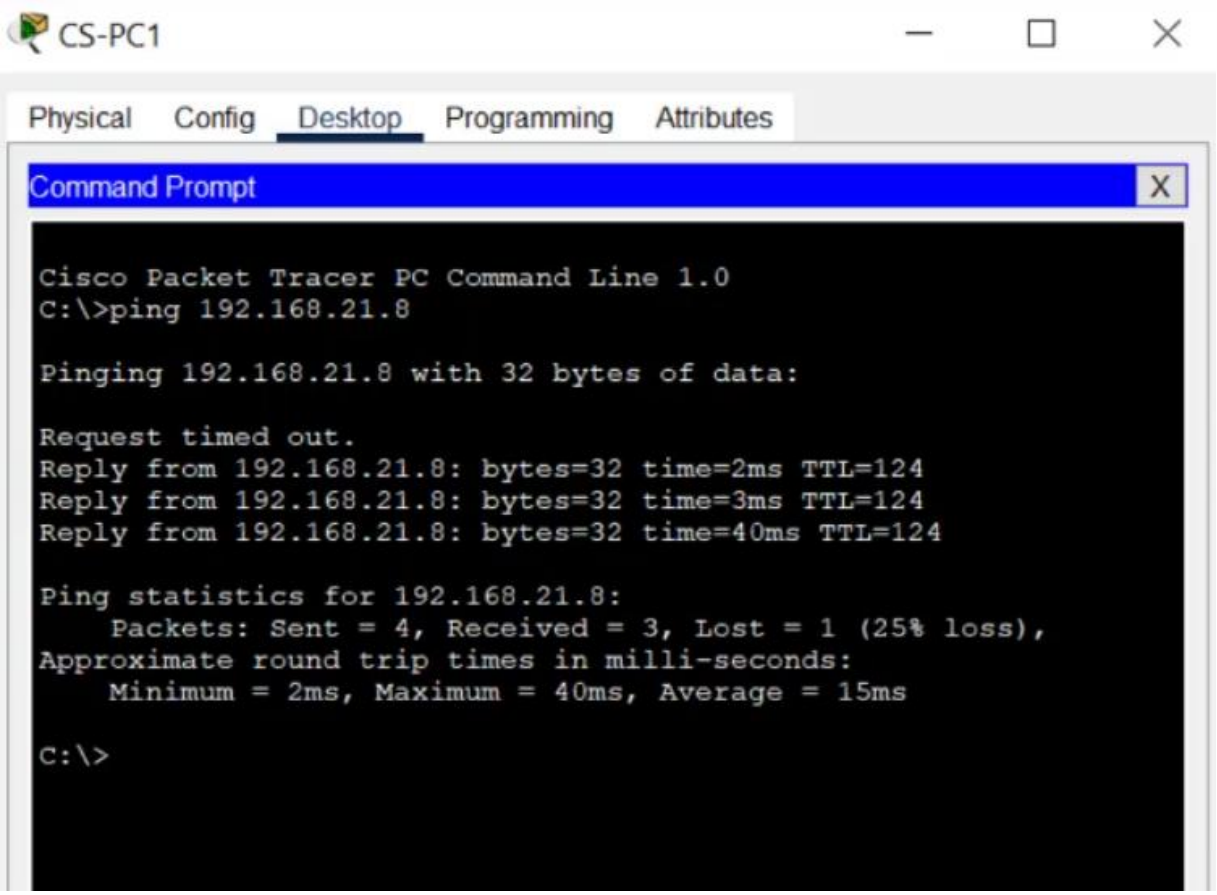
Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.044	IT.SW	IT PC1	OSPF
	0.044	IT.SW	IT Printer1	OSPF
	0.044	IT.SW	IT IP Phone1	OSPF
	0.044	IT.SW	IT IP Phone2	OSPF
	0.044	IT.SW	IT Access Point	OSPF
	0.044	IT.SW	HQ-MLSW1	OSPF
	0.045	IT Access Point	IT Tablet PC	OSPF
	0.045	IT Access Point	IT Laptop	OSPF
	0.045	IT IP Phone1	IT PC2	OSPF
	0.045	IT IP Phone2	IT PC3	OSPF
	0.045	IT Access Point	IT Smartphone	OSPF
	0.045	HQ-MLSW1	HR-SW	OSPF
	0.045	HQ-MLSW1	CS-SW	OSPF
	0.045	HQ-MLSW1	MK-SW	OSPF
	0.045	HQ-MLSW1	L.M.-SW	OSPF
	0.045	HQ-MLSW1	HQ-VoIP Router	OSPF
Visible	0.047	--	ServerSide-Router	OSPF

Reset Simulation ☑ Constant Delay Capture

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	HR PC2	DHCP SE...	ICMP		0.000	N	0	(edit)	(delete)

- **Internet access via NAT**

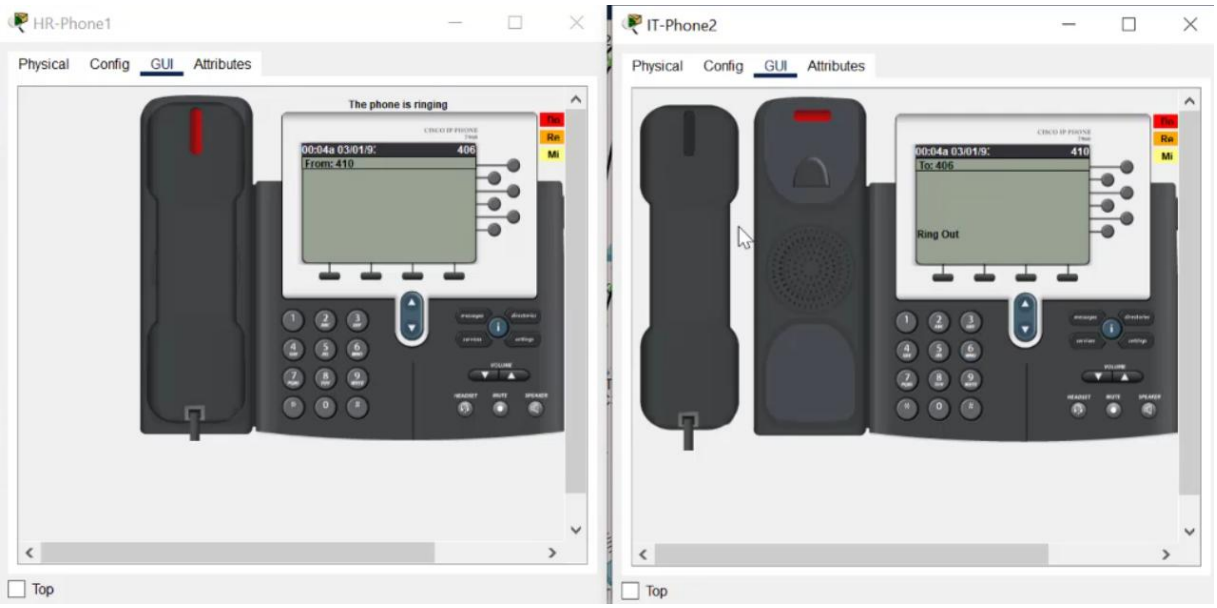


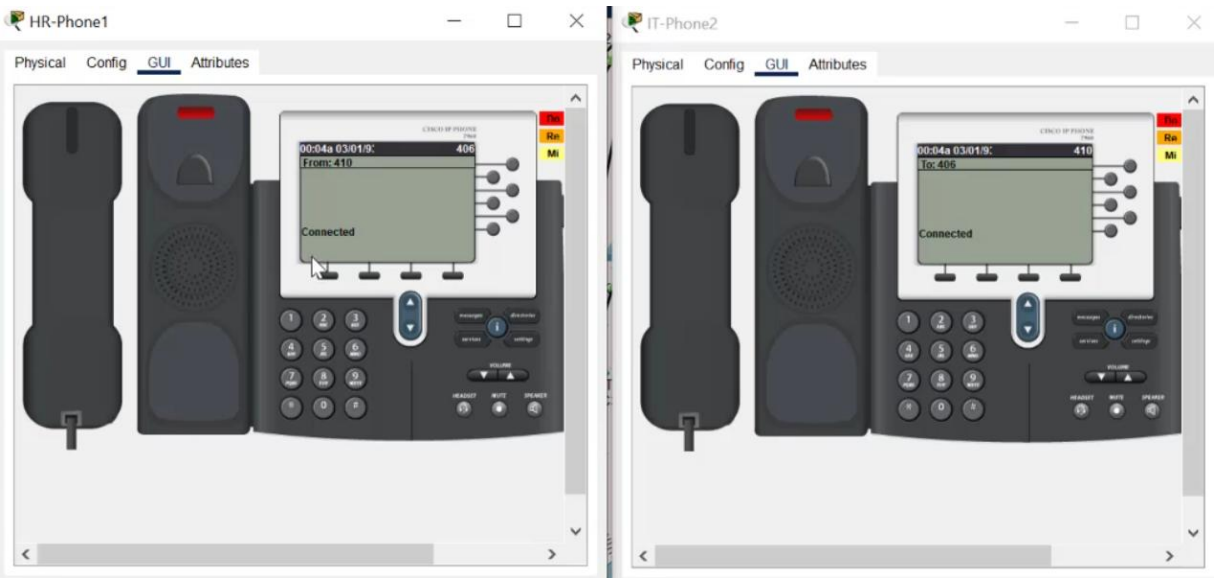
It shows that after the implementation of NAT it is connecting by passing through the router on which NAT implemented.

```
HQ-Router(config)#do sh ip nat translations
Pro  Inside global      Inside local        Outside local
Outside global
icmp 190.200.100.5:1    192.168.20.75:1     192.168.21.8:1
192.168.21.8:1
icmp 190.200.100.5:2    192.168.20.75:2     192.168.21.8:2
192.168.21.8:2
icmp 190.200.100.5:3    192.168.20.75:3     192.168.21.8:3
192.168.21.8:3
icmp 190.200.100.5:4    192.168.20.75:4     192.168.21.8:4
192.168.21.8:4

HQ-Router(config)#
```

Showing that the NAT configured successfully.

- **VoIP call testing**





- **Achievements**

The network operated successfully as per the design requirements. All departments were able to communicate internally and externally, servers were accessible through the VPN, VoIP calls were successful, and redundancy ensured high availability.

- Inter-VLAN communication verified
- OSPF routing tables populated correctly
- DHCP and DNS services operational
- Secure VPN tunnel established

Chapter 5: Conclusion

5.1 Learning Outcomes

This project enhanced understanding of enterprise network design, VLAN segmentation, routing protocols, security implementation, and WAN technologies.

5.2 Challenges Faced

- Complex IP addressing and subnetting
- VPN and NAT configuration troubleshooting
- VoIP integration

These challenges were resolved through systematic testing and verification.

5.3 Future Improvements

- Implementation of QoS for voice traffic
- Network monitoring using SNMP
- IPv6 deployment
- Enhanced firewall policies

5.4 References

- Cisco Networking Academy – Routing and Switching Essentials
- Cisco Packet Tracer Documentation

Course: Computer Networks & Data Communication (CNDC)
Tool Used: Cisco Packet Tracer